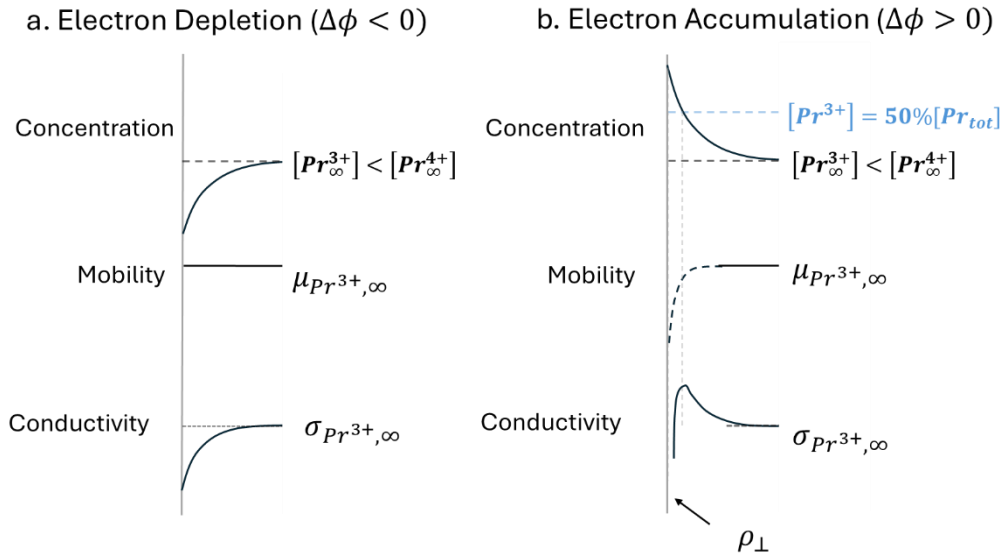
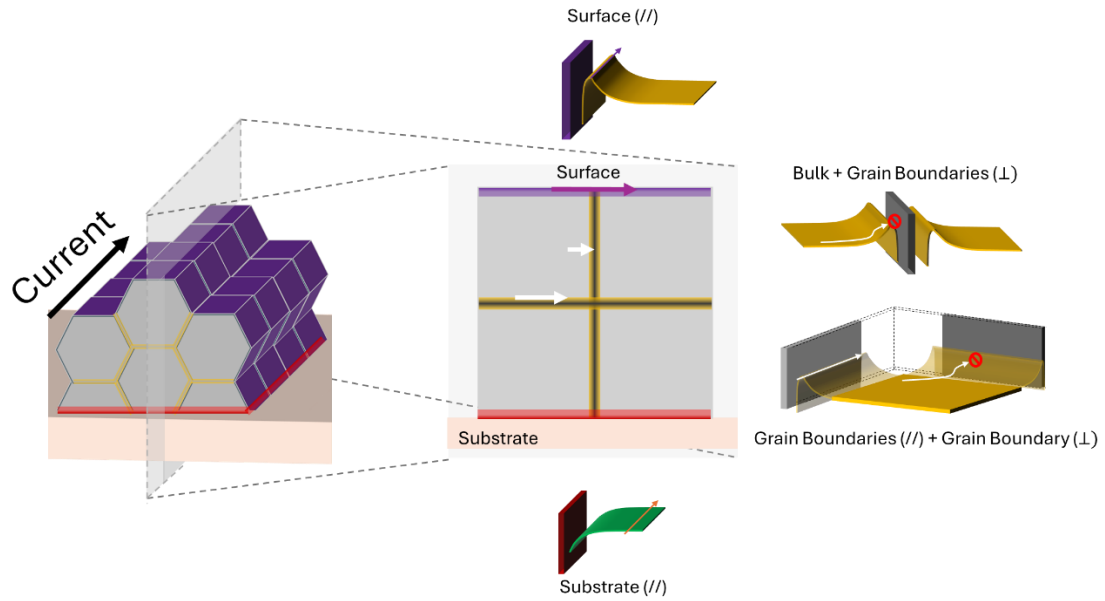
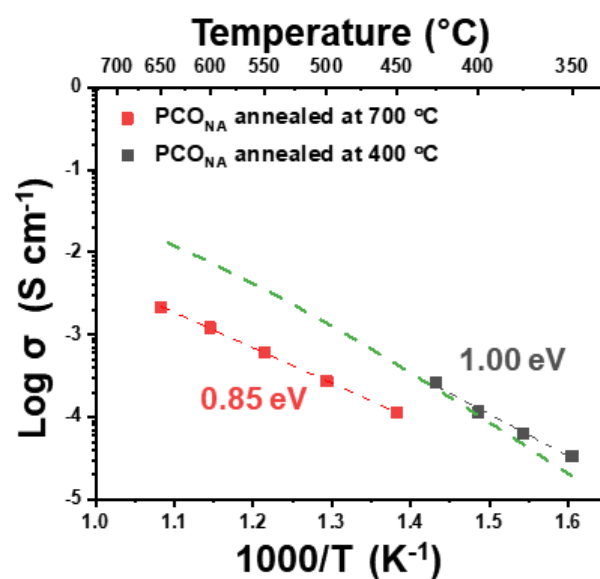




Supplementary Fig. 6 Illustration of space charge model for the case of a. positive and b. negative space charge potential and its impact on the spatial distribution in space charge region of Pr^{3+} concentration ($[Pr^{3+}]$), effective mobility $\mu_{Pr, effective} \sim [Pr^{3+}](x)(1 - [Pr^{3+}](x))\mu_{Pr}$ and expected conductivity (σ).



Supplementary Fig. 7 3D visualization of nanowire composed of multiple nanoscale grains. (left image) cross section of a nanowire – (center image) along nanowire. Purple shaded and gray areas surrounded by yellow shaded areas, represent the surface in contact with a base and grain boundaries where $\Delta\phi > 0$, causing an anisotropic profile of $[Pr^{3+}](x)$ enabling parallel conduction to the surface ($//$), while causing resistive conduction across the grain boundaries ($\perp$). Red shaded areas at the bottom in contact with substrate represent case of nanowire in contact with acidic substrate (i.e. Al_2O_3), expected to cause $\Delta\phi < 0$ and therefore electron depletion.



Supplementary Fig. 8 Arrhenius plots of in-plane conductivity of PCO_{NA} fabricated on Al_2O_3 substrates annealed at 400 $^{\circ}\text{C}$ and 700 $^{\circ}\text{C}$, respectively. The green dashed line indicates the conductivity behavior of bulk PCO_{20} calculated with assistance of defect chemical modelling.